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An Agentic AI System for Automated Pharmacogenomic Recommendation Generation

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Abstract

Pharmacogenomic guidelines are essential for tailoring drug therapy to individual genetic profiles, but current curation workflows are manual, resource-intensive, time-bound, and limited in coverage. We introduce an agentic AI system for automated, scalable generation of CPIC-style recommendations using large language models (LLMs) guided by structured evidence. Our modular pipeline retrieves and processes full-text biomedical literature and FDA drug labels, extracts clinically relevant entities with high accuracy (91.9% across 22 articles), aggregates findings across studies, and generates phenotype-specific dosing recommendations for gene–drug pairs. In expert evaluations of 24 random recommendations, our system significantly outperformed leading LLM baselines (GPT-5, Claude, Grok) in clinical clarity and guideline concordance. These results demonstrate the feasibility of using evidence-grounded, agentic AI for end-to-end pharmacogenomic evidence synthesis, offering a path toward broader population coverage, faster updates, and more consistent and explainable decision support.

Introduction

Pharmacogenomics (PGx) aims to tailor drug therapy based on individual genetic variation, improving both efficacy and safety. To support clinical adoption, expert consortia such as CPIC¹ (Clinical Pharmacogenetics Implementation Consortium, now part of ClinPGx) and DPWG have developed evidence-based guidelines that map genetic variants to dosing adjustments or drug selection rules. These guidelines are widely used in translational research and precision medicine but suffer from two major limitations: (1) the high cost and slow pace of manual curation, and (2) limited coverage of gene–drug pairs and population contexts. As a result, many actionable variants remain unaddressed, and important guideline updates are often delayed by years.

In this work, we focus not on patient-level treatment recommendations but on the automation of guideline-level evidence synthesis, generating structured drafts that can support expert panels, hospital systems, and clinical laboratories in updating and maintaining pharmacogenomic guidance at scale.

A major bottleneck is that current workflows rely on extensive expert involvement for every stage of guideline creation from identifying relevant literature to extracting evidence and formulating recommendations. Even large curated databases such as PharmGKB (now part of ClinPGx) fall short of full coverage: their annotations are manually created and primarily designed for human readability rather than machine interpretation. As a result, integration into automated reasoning pipelines remains challenging, and updates may lag behind the primary literature². Prior efforts such as PGxMine attempted to automate annotation pipelines for PharmGKB, but these remained focused on entity-level extraction without translation into dosing guidance³.

At the same time, large language models (LLMs) have shown increasing promise in biomedical applications, including question answering, summarization, and knowledge extraction. Recent studies have explored the use of LLMs to support or even generate pharmacogenomic recommendations by leveraging structured knowledge bases^{4,5} or free-text literature⁶. In parallel, tools such as PubTator3⁷ have improved normalization of biomedical entities, enabling more reliable downstream reasoning across heterogeneous corpora. These approaches suggest that LLMs, when properly guided and grounded in context, may be capable of capturing complex gene–drug–phenotype relationships, identifying dosing implications, and mimicking the structure and tone of CPIC-like guidelines.

Recent advances in artificial intelligence have introduced a new class of systems known as "agentic AI" — models that go beyond passive information retrieval to actively perceive, reason, and act within dynamic clinical contexts^{8,9}. Similar frameworks have also been applied in drug development and due diligence, for example, LLM-based agent systems for competitive landscape mapping of drug assets¹⁰, underscoring the broader utility of agentic workflows in biomedicine. These agentic systems operate through a perceive–reason–act loop, enabling them to analyze unstructured data, formulate task plans, and autonomously execute actions such as generating clinical summaries or initiating decision support⁸. In healthcare settings, agentic AI has been shown to reduce administrative burdens, enhance diagnostic workflows, and dynamically adapt to changing clinical states — reducing hospital readmissions and improving triage accuracy⁹.

Agentic frameworks are also gaining traction in quantitative clinical pharmacology and translational sciences, where autonomous agents orchestrate multi-step tasks including PK/PD modeling, study simulation, and regulatory document generation¹¹. This multi-agent paradigm, often described as "agentic workflows," enables distributed automation across drug development pipelines, providing both scalability and traceability. In this context, pharmacogenomics stands out as a natural candidate for agentic automation, due to the repetitive, evidence-driven nature of guideline development. By deploying structured agentic reasoning, it becomes possible to automatically identify gene–drug relationships, assess supporting literature, and draft CPIC-style recommendations that are explainable, reproducible, and continuously updatable while also maintaining identifiable distinction from manually curated recommendations.

Our initial attempts followed a traditional path: training supervised models to extract structured annotations from literature, mimicking the format of PharmGKB. However, we found that this approach faced serious challenges: inconsistent formatting in PDFs, limited data coverage, normalization issues for genetic variants, and the high cost of gold-standard labels. Importantly, even if perfect entity extraction were achieved, translating raw evidence into coherent, guideline-style recommendations remained a non-trivial task.

Our system is designed to complement existing expert consortia such as CPIC and DPWG, not replace them. By accelerating evidence synthesis and providing transparent, traceable drafts, it aims to support expert reviewers and enable faster, reproducible updates to clinical guidelines.

We therefore propose a generation-first approach, in which LLMs are used directly to synthesize guideline-like recommendations from raw biomedical literature. Instead of replicating existing

curation schemas, we build a hybrid pipeline that retrieves relevant documents, identifies supporting evidence using semantic search, and prompts an LLM to generate CPIC-style recommendations, including dosage adjustments and rationale. This system can operate both in de novo mode — proposing guidelines for previously uncurated gene–drug pairs — and in updater mode, suggesting revisions for outdated guidelines, for example when new variants are identified in an underrepresented population.

In this paper, we describe the architecture and evaluation of our system. We benchmark our method using expert-validated recommendations, compare several prompting and retrieval strategies, and analyze limitations. We also position our work within recent efforts to benchmark and specialize LLMs for pharmacogenomics^{6,12} and discuss its implications for scalable guideline development in precision medicine.

Results

System Overview

To enable automated generation of pharmacogenomic recommendations, we developed a modular system that processes literature and drug label evidence into structured, guideline-style outputs. The PGxAI-Recommender architecture comprises three stages: acquisition of relevant texts, LLM-based extraction of evidence, and recommendation generation based on structured input (**Figure 1, Supplementary Figure 1**).

This design reflects the culmination of iterative experimentation with multiple prompting strategies, knowledge injection methods, and evaluation setups. Early trials showed that even state-of-the-art LLMs, when prompted solely with a gene–drug pair, frequently returned vague or generic outputs, often omitting critical dosing adjustments. Simply adding all relevant publications to a single prompt is not feasible, as the complete body of evidence for a given gene–drug pair exceeds the available context window of current LLMs. Moreover, an LLM with internet or web-search access cannot reproduce the systematic, curated evidence aggregation performed by guideline authors: ad hoc retrieval returns a limited and opportunistic subset of publications rather than a comprehensive and structured synthesis. These limitations motivated a more structured, evidence-aware pipeline tailored to the complexities of pharmacogenomic decision-making.

Evaluation of Extraction Accuracy

We manually validated the outputs of the LLM-based evidence extraction module on 22 pharmacogenomic articles. Three independent human annotators evaluated the correctness of the

extracted answers across 35 predefined binary fields per article. In total, 2310 binary judgments were collected across annotators and articles.

The model achieved an overall accuracy of 91.9%, defined as the proportion of fields marked as correct by the annotators. Errors primarily reflected incorrect or imprecise extractions, including occasional over-specification.

Structured factual fields such as Study design, Genotyping method, and Sample size achieved near-perfect performance, while interpretive or composite fields, including Statistical significance and strength and Phenotypes reported, showed relatively lower accuracy, with the minimum observed for Genotype distribution (56%) (**Figure 2A**).

Per-article accuracy was consistently high (**Figure 2B**), with most articles exceeding 90% correctness across all evaluated fields and only a few outliers falling slightly below 85%.

Inter-rater reliability among annotators, quantified using pairwise Cohen's κ (0.19, 0.23, 0.52) and Fleiss' κ (0.26), indicated substantial variability across reviewers. This variability is partly attributable to the strong imbalance between correct and incorrect labels and to differences in expert interpretation for a subset of fields, where reasonable judgments may diverge.

To contextualize model performance within expert variability, we examined how often extracted fields were accepted by expert annotators. Annotator-majority agreement ranged from 88.3% to 98.1% (**Figure 2C**). For the model, 94.7% of extracted fields were judged correct by the majority of expert annotators, indicating that model outputs are generally consistent with prevailing expert assessments.

Evaluation of Recommendation Accuracy

To assess the clinical quality of generated pharmacogenomic recommendations, we conducted a blinded expert review of outputs from five models. Each recommendation was rated on a 0–10 scale based on alignment with known CPIC guidelines, clinical clarity, and dosing appropriateness. PGxAI-Recommendier achieved the highest average expert score (mean 9.0), compared to baseline models with mean scores ranging from 6.2 to 7.8, and outperformed all baseline LLMs in pairwise statistical comparisons ($p < 0.01$; **Figure 3B**).

To further contextualize the score-based evaluation, we conducted a pairwise guideline-conformance assessment against CPIC reference recommendations. In this analysis, we derived a relative ranking of models based on expert preferences. PGxAI-Recommendier achieved the highest observed pairwise win rate (0.83), whereas baseline LLMs exhibited lower win rates

ranging from 0.17 to 0.78, indicating stronger alignment of PGxAI-Recommend outputs with CPIC guidance (**Supplementary Figure 2**).

A notable observation from the baseline comparison was that GPT-5, despite being a more advanced model, did not improve recommendation quality and in several cases produced less guideline-concordant outputs. GPT-5 occasionally generated detailed dosing instructions that contradicted CPIC guidance. In scenarios where all other models correctly returned “No recommendation,” GPT-5 produced full therapeutic regimens with specific milligram dosing and administration schedules, even though such information was neither present in the evidence corpus nor requested by the task. In other instances, GPT-5 returned overly narrow dose values (for example, a single fixed efavirenz dose) where CPIC requires expressing a dose range. These deviations represent over-generation rather than guideline-aligned reasoning and likely contributed to its lower expert scores relative to earlier GPT models.

Three representative cases highlight the model’s strengths.

Efavirenz – *CYP2B6* (Multiple Phenotypes).

CPIC provides nuanced, phenotype-specific guidance for efavirenz, recommending dose reductions for intermediate and poor metabolizers and careful monitoring or escalation for ultrarapid metabolizers. The PGxAI-Recommend consistently produced accurate and differentiated outputs across all phenotypes. For intermediate and poor metabolizers, it suggested the correct reduced dose range (200–400 mg), while most baseline models either defaulted to the standard 600 mg or gave only partial adjustments. For ultrarapid metabolizers, our system appropriately suggested standard dosing with optional escalation (up to 800 mg) under monitoring, striking a balance between clinical flexibility and evidence caution. Other models frequently omitted dose adjustments or proposed incorrect increases (e.g., 900 mg), diverging from established recommendations. This case illustrates the system’s ability to align with guideline logic even in edge phenotypes requiring therapeutic nuance.

Fluorouracil – *DPYD* (Normal and Poor Metabolizers).

For normal metabolizers, CPIC advises standard dosing without modification. Most baseline models, however, added unnecessary regimen details (e.g., “400–600 mg/m²/day” or complex bolus + infusion schedules), which could mislead clinicians and complicate justification during adverse events. In contrast, the PGxAI-Recommend returned the concise and guideline-aligned statement, standard dosing with no changes, avoiding extraneous detail. For poor metabolizers,

CPIC guidance is clear: avoid fluorouracil or related prodrugs due to the risk of severe, potentially life-threatening toxicity. If no alternatives exist, only a highly reduced dose with intensive therapeutic drug monitoring may be considered. The PGxAI-Recommender precisely captured this “no safe use” stance, stating that fluorouracil should be avoided and offering no ambiguous fallback. Other models tended to hedge, some suggested 25–50% of the standard dose or added vague regimen-specific caveats, which could expose patients to undue risk. Our system’s categorical rejection aligns most closely with clinical best practice, reducing the chance of misinterpretation in critical care settings.

Atorvastatin – *SLCO1B1* (Poor Function).

CPIC suggests avoiding high-dose atorvastatin and considering alternatives such as pravastatin. The PGxAI-Recommender explicitly recommended ≤ 20 mg/day or switching to a safer statin, whereas other models proposed standard dose ranges (e.g., 10–80 mg/day) or omitted alternatives altogether. Our system framed dose reduction in the context of broader therapeutic strategy, yielding a more actionable and complete recommendation.

These findings highlight the value of evidence-aware prompting and structured input design in generating high-quality, guideline-concordant pharmacogenomic recommendations.

Discussion

Our evaluation demonstrates that LLMs can reliably extract structured evidence when provided with curated, task-specific context. The extraction module achieved an overall field-level correctness rate of 91.9% across 22 pharmacogenomic studies, as assessed by independent expert annotators. Agreement with the majority expert consensus was 94.7%, indicating that the majority of extracted fields were judged acceptable by domain experts. At the same time, inter-rater reliability metrics (Cohen’s and Fleiss’ κ) revealed variability among expert judgments. This variability reflects both the imbalance between correct and incorrect labels and differences in expert interpretation for a subset of fields, rather than systematic extraction errors. Together, these results indicate that while expert consensus provides a useful operational reference, some degree of disagreement is intrinsic to the task and should be taken into account when interpreting extraction performance.

In our expert evaluation, the PGxAI-Recommender consistently outperformed all baseline models in generating pharmacogenomic recommendations aligned with CPIC guidelines. The system

achieved significantly higher expert scores across a diverse set of gene–drug–phenotype cases, particularly excelling in scenarios that required non-standard clinical actions. In phenotypes such as *CYP2B6* poor and ultrarapid metabolizers for efavirenz, or *SLCO1B1* poor function for atorvastatin, our model uniquely captured the need for dose reduction, therapeutic monitoring, or alternative agents—clinical subtleties that baseline LLMs frequently missed or expressed imprecisely. In addition to score-based expert evaluation, we performed a pairwise CPIC-guideline conformance assessment to explicitly examine relative model preference. This analysis consistently ranked PGxAI-Recommendier above baseline LLMs, reinforcing its stronger alignment with established pharmacogenomic guidelines. These results underscore the importance of grounding generative models in structured, context-specific evidence to enable guideline-concordant outputs that are both actionable and safe for clinical decision support.

A key difference between the PGxAI-Recommendier and baseline models such as GPT-4.1 lies in how each system uses evidence. GPT-4.1 relies solely on a single prompt applied to an isolated gene–drug–phenotype description, without access to aggregated multi-study evidence or FDA label information. In contrast, our system integrates structured inputs assembled across several upstream stages, including study-level annotations, multi-document evidence summaries, FDA excerpts, and a qualitative evidence level. This consolidated evidence block provides explicit grounding signals, numeric results, consistency across studies, phenotype definitions, and regulatory context, that the generator is required to use when formulating the final recommendation. This architecture reduces unsupported assertions, improves dose-range fidelity, and enables the system to capture guideline-specific subtleties (such as recommending alternative agents or monitoring strategies) that GPT-4.1 frequently omitted or expressed incompletely. We believe this deeper evidence integration explains the observed performance gap in expert ratings. Importantly, the observed performance gains reflect the use of structured, guideline-relevant context and prompt design rather than any change to the underlying language models. PGxAI-Recommendier does not involve model training or fine-tuning. All models were evaluated in an inference-only setting without internet access.

Together, these findings highlight the utility of evidence-aware prompting for achieving guideline-concordant outputs, and support the use of structured knowledge to enhance reliability in clinical NLP applications.

Beyond accuracy, our system demonstrates the potential for broader coverage and more up-to-date evidence compared to existing expert-curated resources. As summarized in Table 1, the CPIC corpus currently includes 316 gene–drug pairs and roughly 8,000 cited articles, reflecting guidelines published between 2014 and 2025. In contrast, the proposed automated framework can identify up to 573 gene–drug pairs and retrieve nearly 20,000 relevant pharmacogenomic publications indexed in PubMed (1966–2025), supporting the generation of continuously updated, literature-grounded recommendations that align with the evolving evidence base.

Although the extraction and recommendation evaluations relied on a relatively small set of manually reviewed articles (22) and recommendations (24), these analyses were intended as a feasibility assessment rather than a full validation study. The system itself is built to operate at much larger scale, and future work will include validation on substantially expanded and more heterogeneous evaluation sets as additional expert-curated data become available.

Access to full-text articles represents an additional practical limitation. Many pharmacogenomic studies are distributed across heterogeneous publisher platforms with inconsistent open-access availability. To maximize coverage, the retrieval module attempts multiple pathways, including PMC, DOI-redirectioned publisher pages, and applies PDF parsing and OCR-based extraction when needed. Nevertheless, studies available only behind paywalls remain inaccessible to the current pipeline, which may introduce coverage gaps or bias toward open-access literature. Future iterations could incorporate licensed full-text sources or publisher partnerships where permitted.

Although several domain-specific fine-tuned models exist, including BioMedLM¹⁶, BioGPT¹⁷, and Me-LLaMA¹⁸ variants trained on PubMed or clinical corpora, we did not include them in the present evaluation for two main reasons. First, in the context of multi-step evidence synthesis — which requires long-context handling, and reliable adherence to structured prompts — recent general-purpose LLMs substantially outperform domain-specific models. Second, preliminary exploratory tests (conducted outside the scope of this manuscript) indicated that while domain-adapted models performed adequately on narrow extraction tasks, they were markedly less reliable in tasks requiring multi-document integration and structured recommendation generation. Given these limitations, and considering the rapid progression of frontier LLM capabilities, we focused the formal evaluation on state-of-the-art general-purpose models. Future work may revisit domain-specific architectures as they continue to mature and as larger, more comprehensive biomedical training corpora become available.

Panel-based guideline development remains the gold standard, and our system is intended to support, not replace, expert-driven curation by accelerating evidence synthesis and producing traceable drafts for review. Several community initiatives already contribute to systematizing pharmacogenomic knowledge at scale, including automated extraction pipelines and expert-curated resources. PGxMine, for example, leverages large-scale text mining across PubMed to surface gene–drug–variant signals, offering broad coverage but necessarily limited to shallow associations rather than full study-level evidence. Complementary resources such as PharmGKB and its successor ClinPGx remain authoritative repositories built through expert review, standardized annotation, and consensus-based evidence grading. The framework described here is designed to operate alongside these efforts: it uses a multi-step, evidence-synthesis workflow to integrate findings across studies and generate guideline-formatted drafts, providing a bridge between automated discovery pipelines and expert curation while remaining aligned with established community standards.

While this study focuses on gene–drug pairs, real-world pharmacogenomic decision-making also depends on broader context, including drug–drug interactions, phenoconversion, prior treatments, polygenic influences, patient age, comorbidities, and disease-specific factors. The current system does not model these factors directly, but its modular evidence schema is compatible with incorporating additional structured inputs. Extending the pipeline to capture these modifiers represents an important direction for future work.

Beyond these clinical modifiers, the capabilities of the described approach are not limited to CPIC-style gene–drug recommendations. The modular structure and agentic controller can be adapted to alternative pharmacogenomic standards, including DPWG or others. It can also be extended beyond pharmacogenomic guidelines. In principle, the same architecture can incorporate additional molecular modalities, such as transcriptomic activity, metabolomic signatures, or other omics-derived features, whenever sufficient evidence data are available. Extending the system to these broader contexts represents an important direction for future development.

The current report represents a preliminary estimate rather than a fully validated corpus, however, it illustrates the scalability of our automated pipeline and its capacity to incorporate emerging evidence from the latest literature in conjunction with expert review. This expanded scope supports applications beyond traditional curation, including faster updates, context-specific recommendations, and broader population coverage.

Methods

End-to-End System for Pharmacogenomic Recommendation Generation

This section describes the full architecture and operational flow of the PGxAI-Recommend system as designed for real-world deployment, capable of retrieving, processing, and integrating new evidence continuously. The process is composed of the following steps:

Drug–Gene Pair Selection

The process begins by carefully selecting a specific drug and gene pair of interest. This selection is crucial because it defines the scope of the literature review and data sources that will be queried in the subsequent steps. Understanding the interaction between a particular drug and genetic variations is a fundamental aspect of precision medicine.

These studies are highly important as they provide insights into how genetic differences among individuals can influence the efficacy and safety of medications. For clinicians, having detailed knowledge about multiplex drug–gene interactions enables more informed decision-making when prescribing treatments. This helps to optimize therapeutic outcomes, minimize adverse drug reactions, and tailor treatments to the genetic profile of each patient.

Ultimately, investigating drug–gene pairs supports the development of precision medicine strategies, allowing healthcare providers to move beyond the traditional “one-size-fits-all” approach and towards more personalized, effective, and safer therapies.

Literature Retrieval

The system searches PubMed database (1966–2025) through the Entrez Programming Utilities¹³. Structured queries are constructed using the following keywords: gene name AND drug name, searched across Title/Abstract, MeSH Terms, and All Fields.

The search was designed to identify peer-reviewed articles, clinical studies, and reviews describing gene–drug associations, pharmacokinetic or pharmacodynamic effects, and genotype-guided dosing outcomes.. For each retrieved PMID, an attempt is made to download the article in XML format from PubMed Central (PMC).

If the XML is successfully retrieved, the system parses and extracts relevant textual content, carefully removing non-informative XML tags to focus on the essential scientific information.

For PMIDs where XML-format articles are not available via PubMed, retrieving full-text content poses a significant challenge due to inconsistent access across publishers and repositories. To

address this, we implemented a robust retrieval module that systematically searches for full-text PDF versions across a variety of open-access sources, including publisher websites, institutional repositories, and document aggregators: PMC, DOI redirect, Springer, Wiley Online Library, Frontiers. If a PDF is successfully located, it is processed using a dedicated parsing engine based on the pypdf Python library to extract embedded text.

In cases where the PDF contains scanned images or non-extractable text, the system applies an OCR fallback pipeline using pdf2image to render the pages as images and pytesseract to extract text content. All extracted texts, whether from XML, abstract, PDF, or OCR sources, are normalized, cleaned, and stored in a centralized database indexed by PMID to ensure traceability and reproducibility during downstream analysis.

Annotation via Large Language Model (LLM)

To support the automation of clinical interpretation, each retrieved full-text document is processed by a large language model (LLM) using a custom-designed prompt tailored for pharmacogenomics. This step uses GPT-4.1 (OpenAI API, temperature = 1e-5), the most current model available at the time of system development. The primary objective is to identify and extract clinically relevant entities and relationships that would typically be assessed manually by a domain expert.

The LLM identifies key elements such as:

- Drug name
- Gene and specific allele(s)
- Pharmacokinetic/pharmacodynamic (PK/PD) parameters
- Phenotypic consequences (e.g., metabolizer status)
- Comparative analysis of allelic variants
- Study-level associations and outcomes

This step enables the systematic extraction of high-value information from unstructured texts, regardless of document format or writing style. Example in Supplementary Table 1. The resulting structured annotations are stored in the database and serve as the foundation for generating precise, guideline-ready therapeutic recommendations.

Evaluation results and performance metrics are presented in a later section of this paper.

Evidence Aggregation Across Multiple Studies

To improve consistency and reliability of downstream recommendations, we implemented an intermediate step that aggregates structured findings across multiple studies for each gene–drug pair. For every set of extracted annotations (grouped by PubMed ID), the system clusters evidence by allele (e.g., *2, *4, rsIDs) and organizes results into clinical categories: enzyme activity, dose adjustment, efficacy, toxicity, and PK/PD outcomes. A language model then summarizes the findings into unified evidence records, modeled after CPIC-style evidence tables. These include:

- A brief description of the main effect (e.g., “reduced activity in carriers of *2A”)
- Supporting vs. conflicting references by PMID
- Type of evidence (e.g., Clinical, In vitro)
- Heuristic level of evidence (High / Moderate / Weak)

For each aggregated gene–drug–phenotype evidence block, the system assigns a qualitative level of evidence (High, Moderate, or Weak) using an LLM-driven assessment. Rather than imposing fixed numerical thresholds, we follow the principles articulated in the foundational CPIC evidence framework¹⁴, which emphasize reproducibility, study quality, and consistency across independent investigations.

The LLM receives a consolidated structured summary of all extracted studies and applies the following criteria:

- High: consistent results from multiple well-designed studies, including reproducible associations, clear directionality of effect, appropriate statistical support, and concordant clinical or pharmacokinetic outcomes.
- Moderate: findings supported by multiple studies but with limitations such as smaller sample sizes, partially inconsistent results, indirectness, or methodological constraints that reduce confidence but still indicate a likely effect.
- Weak: insufficient basis for determining a reliable effect because evidence is sparse, contradictory, underpowered, methodologically limited, or lacks key components in the chain of inference.

These criteria are encoded directly in the LLM prompt, ensuring consistent reasoning across cases and avoiding rigid numerical cutoffs that may not generalize across heterogeneous study designs. While the process is heuristic and LLM-mediated, it follows established qualitative frameworks used in pharmacogenomic guideline development and provides a transparent basis for downstream

recommendation generation. An example output from this step is provided in Supplementary Table 2.

Preparing of Additional Data

To incorporate regulatory guidance, the system retrieves FDA¹⁵ drug label PDFs directly from the official Drugs@FDA database and the FDA Label Search (NCTR) portal (U.S. Food and Drug Administration, accessed June 2025). Extracted label text is processed with GPT-4.1 using a custom prompt to extract key prescribing information, including dosage, pharmacogenetic considerations, contraindications, and relevant warnings. The structured output is stored in the database and used alongside literature-derived evidence during recommendation generation. An example output from this step is provided in Supplementary Table 3.

Recommendation Generation

To generate structured clinical guidance, the system uses a dedicated recommendation module built on GPT-4.1 via the OpenAI API (temperature = 1e-5) with a task-specific prompt.

The model receives the following inputs:

- Aggregated evidence block: consolidated findings from multiple studies (Section 4.1), grouped by allele and outcome type (e.g., activity, dose, efficacy, toxicity, PK/PD).
- FDA drug label excerpts, including dosage and pharmacogenomic considerations.
- Target gene–drug–phenotype triple for which a recommendation is being generated.

The model is explicitly instructed to:

- Decide whether to use the drug for the specified phenotype.
- Suggest a dose (standard, reduced, increased) with numeric values when possible.
- Specify administration frequency (e.g., once daily).
- Assign a recommendation strength: Strong, Moderate, or No recommendation.
- The output follows a strict JSON format and serves as a guideline-style recommendation compatible with CPIC structure.

This step is fully automated, consistent, and scalable—enabling the generation of hundreds of phenotype-specific recommendations with clinical clarity and traceable evidence provenance.

An example of the final structured output is provided in Supplementary Table 4, with the corresponding JSON representation shown in Supplementary Note 1.

Storing and Continuous Updates Based on New Publications

All structured annotations, aggregated evidence tables, and generated recommendations are stored in a centralized database, indexed by PMID and gene–drug–phenotype triplet, enabling traceability, reproducibility, and scalable querying.

To ensure that recommendations remain up-to-date, the system performs an automated weekly literature scan and incrementally re-processes affected gene–drug. When new data is detected, the same processing pipeline (retrieval → extraction → annotation → structuring → recommendation) is re-run. This automated updating mechanism enables the system to continuously maintain current and comprehensive pharmacogenomic knowledge. As a result, it supports generating high-quality, evidence-based conclusions and recommendations that reflect the latest scientific findings, thereby enhancing clinical decision-making and guideline accuracy over time.

Agentic orchestration

To coordinate the multi-step processing required for each gene–drug pair, the PGxAI- Recommender incorporates a lightweight agentic controller that operates using a perceive–reason–act loop. Instead of executing a fixed sequence of steps, the system exposes each processing module—literature retrieval, document annotation, FDA label parsing, evidence aggregation, and recommendation generation—as an independent callable tool. The controller maintains a persistent state for every case in a dedicated results table, with fields indicating whether each stage (extract, annotate, fda, recs) is pending, completed, or requires re-execution.

At runtime, the controller receives a high-level goal (e.g., “complete processing for case X”) and periodically queries the database to perceive the current state of that case. Based on this state, it reasons about which action is required next: retrieving missing full texts, annotating newly downloaded documents, aggregating available evidence into phenotype-specific Level-of-Evidence summaries, or triggering guideline-style recommendation generation. After each tool invocation, the controller writes updated results and logs back to the database, forming a feedback loop in which subsequent decisions depend entirely on the evolving system state rather than a predetermined pipeline.

This design enables the agent to skip already completed stages, repeat failed or incomplete steps, adapt to heterogeneous document availability, and terminate automatically once all required components have been produced. The architecture therefore supports robust error recovery, deterministic state tracking, and transparent auditability while preserving flexibility in execution

order. A concise version of the controller's instruction prompt is provided in Supplementary Note 2, and a pseudocode representation of the perceive–reason–act workflow is provided in Supplementary Note 3. A schematic overview of the orchestration process is included in Supplementary Figure 1.

Experimental Setup for Evaluation

While the PGxAI-Recommendier is designed for continuous evidence integration from literature and regulatory sources (see "End-to-End System for Pharmacogenomic Recommendation Generation"), the evaluation in this paper was conducted on a fixed set of 17 gene–drug pairs derived from existing CPIC guidelines. Literature used for recommendation generation was selected from the CPIC bibliography to enable direct comparison with reference recommendations. The evaluation of the system was conducted in two distinct stages to ensure both the accuracy of extracted annotations and the clinical relevance of generated guideline recommendations.

Annotation Validation

The first evaluation stage assessed whether the LLM-based annotation module accurately extracted clinically relevant information from primary literature. Three human annotators, including a PGx-trained reviewer, independently evaluated each extracted value and labeled it as correct or incorrect for a random sample of 22 full-text articles. For each article, annotators compared the structured outputs generated by the system with the original text to assess whether key data points, such as gene variants, phenotypic effects, and pharmacokinetic outcomes, were correctly identified and extracted. Importantly, the extraction task requires the model to produce a value for every predefined field in each article, and annotators assess only the correctness of the generated value rather than the presence or absence of the field in the source text. The extracted fields span heterogeneous data types, including quantitative values, categorical descriptors, and identifiers. As a result, classical positive and negative categories are not uniformly applicable across fields in this evaluation. Accordingly, the evaluation focused on completeness and factual accuracy of the annotations as a prerequisite for reliable downstream recommendation generation. Accuracy was calculated as the proportion of extracted fields that annotators judged to be correct. Full expert scoring results are available on Zenodo (DOI: 10.5281/zenodo.17727992).

Recommendation Quality Assessment

To evaluate clinical quality, we conducted a blinded expert review across a representative subset of random 24 recommendations. Selection criteria prioritized diversity across genes, drugs, and

metabolizer phenotypes (e.g., poor, intermediate, ultrarapid), ensuring clinically meaningful variation.

To benchmark overall quality, we compared recommendations produced by the full PGxAI-Recommendier — which integrates structured evidence, regulatory label data, and gene–drug–phenotype logic — against outputs from baseline large language models (GPT-4.1, GPT-4o-mini, GPT-5, GPT-o3 by OpenAI; Claude Opus 4 by Anthropic; and Grok 4 by xAI) prompted under identical conditions but without access to structured evidence or curation logic. The exact prompt templates used for both the baseline condition and the evidence-augmented recommendation module are provided in Supplementary Notes 4 and 5. For clarity, the PGxAI-Recommendier and the “baseline GPT-4.1” condition share the same underlying model; the only difference lies in the structured evidence block.

All outputs were independently evaluated by a clinical pharmacologist (MD, PhD) with formal training in pharmacogenomics. The reviewer applied a standardized 0–10 scoring rubric, where:

0–3 indicated incorrect or clinically misleading content,

4–6 represented partially aligned or incomplete recommendations,

7–8 reflected guideline-concordant outputs with minor omissions, and

9–10 denoted statements indistinguishable from reference CPIC guidance.

The expert remained blinded to model identity throughout the evaluation.

In addition to score-based evaluation, we performed a blinded, randomized pairwise guideline-conformance assessment against CPIC reference recommendations. For each pharmacogenomic case, the expert compared anonymized pairs of model outputs and selected the option more consistent with CPIC guidance, or indicated a tie or that neither option was consistent. To minimize potential order and presentation bias, the order of comparison cases and the left–right positioning of model outputs were randomized. Pairwise outcomes were used to derive relative preference rankings based on decisive comparisons. Pairwise win rates were computed as the proportion of decisive comparisons (wins divided by wins plus losses), with ties excluded from the denominator. Full expert scoring results are available on Zenodo (DOI: 10.5281/zenodo.17727992)

Illustrative examples and reproducibility.

For clarity and reproducibility, we provide representative examples of model outputs at each stage of the workflow. Using the Efavirenz–CYP2B6 pair as a reference case, the supplementary materials illustrate how unstructured literature and regulatory data are transformed into structured

annotations, aggregated evidence tables, drug label summaries, and final guideline-style recommendations (Supplementary Tables 1-4). These examples demonstrate the transparency and traceability of the PGxAI-Recommender process and allow readers to inspect the structure of intermediate outputs in detail.

Analysis and Visualization

Quantitative evaluation results were analyzed using Python-based scientific computing libraries. Pairwise Wilcoxon signed-rank test was performed using the `scipy.stats` module to assess differences in expert scores across models. Data wrangling and metric computation were handled using `pandas` and `numpy`. For visualization, we used `matplotlib` and `seaborn`.

Ethics statement

This study did not involve human participants or human biological materials and did not use identifiable human data; therefore, ethics approval and informed consent were not required.

Data Availability

All data analyzed in this study are derived from publicly available PubMed-indexed literature and FDA drug labels. The expert evaluation data are available at <https://doi.org/10.5281/zenodo.17727992>.

Code Availability

The code used in this study is not publicly available due to proprietary constraints.

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Author Contributions

M.Z. conceived and supervised the study. M.Z. and D.St. defined the research objectives and overall study design. A.S. developed and implemented the agentic AI pipeline and computational infrastructure. D.St. and K.R. performed bioinformatic analyses, data processing, and validation. A.J.M., A.Y., and N.Y. collected source literature, curated datasets, and contributed to data interpretation. D.So. and I.T. prepared the figures and tables. M.Z. drafted the manuscript with critical input from D.St. and I.T. A.G. and M.Z. secured funding, provided resources, and managed project administration. All authors (M.Z., A.S., D.St., A.J.M., D.So., I.T., A.Y., K.R., N.Y., A.G.) reviewed and revised the manuscript for important intellectual content, approved the final version, and agree to be accountable for all aspects of the work.

Competing interests

All authors are employees, contractors, and/or equity holders of PGxAI Inc., which may commercialize technologies described in this manuscript. The authors declare no other competing interests.

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Figure legends

Figure 1. Architecture of the PGxAI-Recommender.

Final system design incorporating structured literature-derived evidence prior to recommendation generation. This evidence-aware pipeline emerged from comparative benchmarking and expert scoring, demonstrating substantial gains over baseline prompting alone.

Figure 2. Accuracy of structured evidence extraction.

(A) Field-level accuracy, averaged across 22 articles and three annotators. Accuracy represents the proportion of extracted fields that annotators judged to be correct. The model achieved high accuracy on structured factual items, with lower scores on interpretive fields. (B) Per-article accuracy, showing the proportion of fields judged correct per article. (C) Agreement with majority expert consensus. For human annotators, bars indicate the proportion of field-level judgments that matched the majority expert label. For the model, the bar indicates the proportion of fields judged correct by the majority of annotators. These values represent descriptive agreement with expert consensus and are not corrected for chance; inter-annotator reliability is assessed separately using Cohen's κ and Fleiss' κ .

Figure 3. Expert Ratings of Guideline-Style Recommendations Across Models.

(A) Overview of the recommendation generation workflow. The PGxAI-Recommender incorporates structured pharmacogenomic evidence prior to generating guideline-style outputs. Baseline models receive only a gene–drug pair and rely on tuned prompts. (B) Average expert scores (0–10) across models. Bars show the mean rating across 24 recommendations per model. Error bars represent 95% confidence intervals. PGxAI-Recommender achieved the highest median rating (9.0/10), outperforming all baseline LLMs. Statistical testing confirmed significant differences across models (Friedman $\chi^2 = 46.71$, $p < 0.001$). Pairwise Wilcoxon tests with Holm correction showed that PGxAI-Recommender significantly outperformed all baseline models (adjusted $p < 0.01$).

Table legends

Table 1. Potential coverage and update frequency of the PGxAI-Recommender compared with CPIC.

ARTICLE IN PRESS

Source	Gene–Drug Pairs	Cited Articles	Update Frequency
CPIC	316	~8,000	Periodic (manual)
PGxAI-Recommender	573	~20,000	Continuous (automated)

PGxAI-Recommender

Gene-Drug Pair
(input)

1. Literature Acquisition

- Metadata from PubMed
- Manuscripts from OA Sources
- Text parsing and normalization

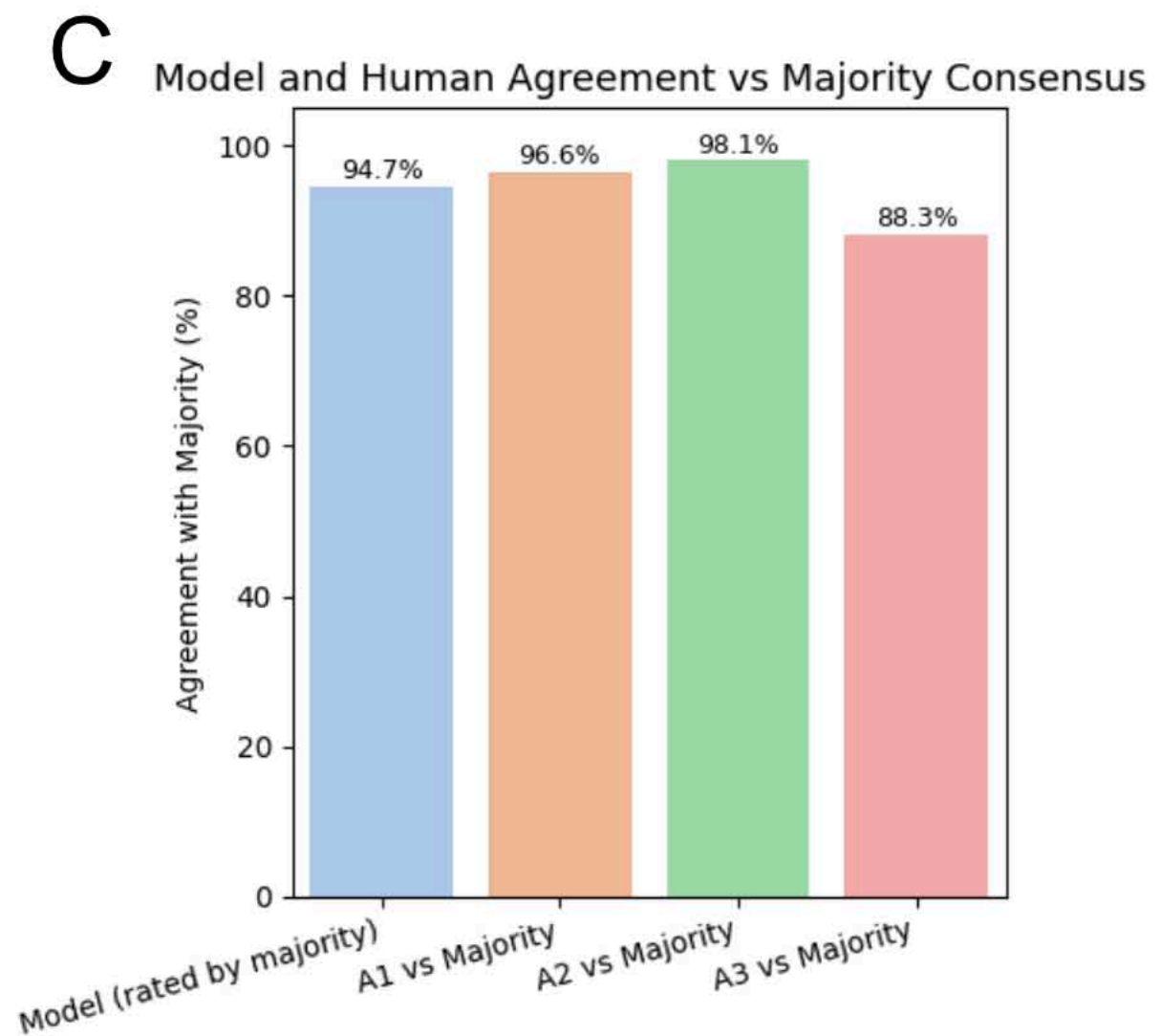
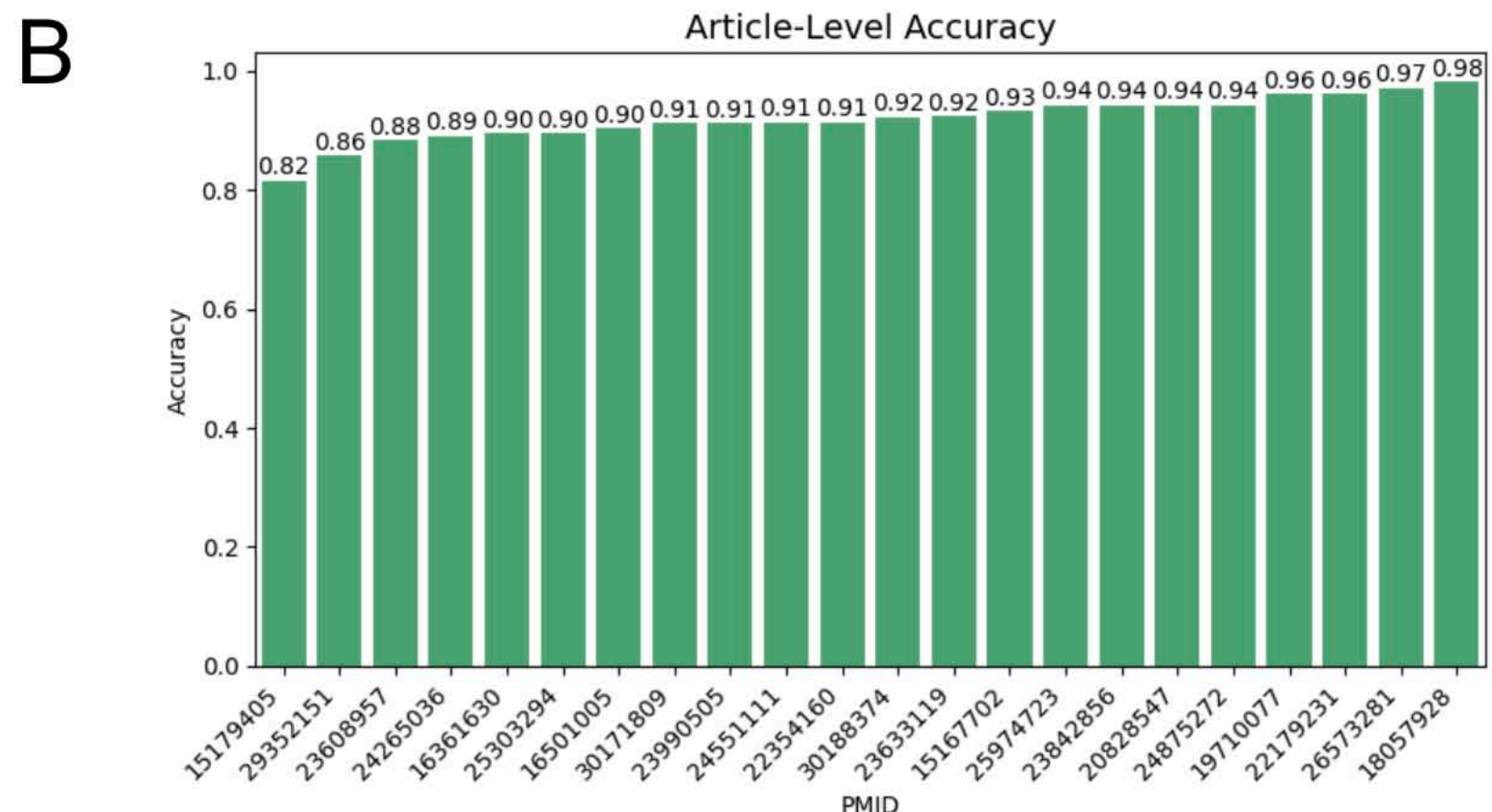
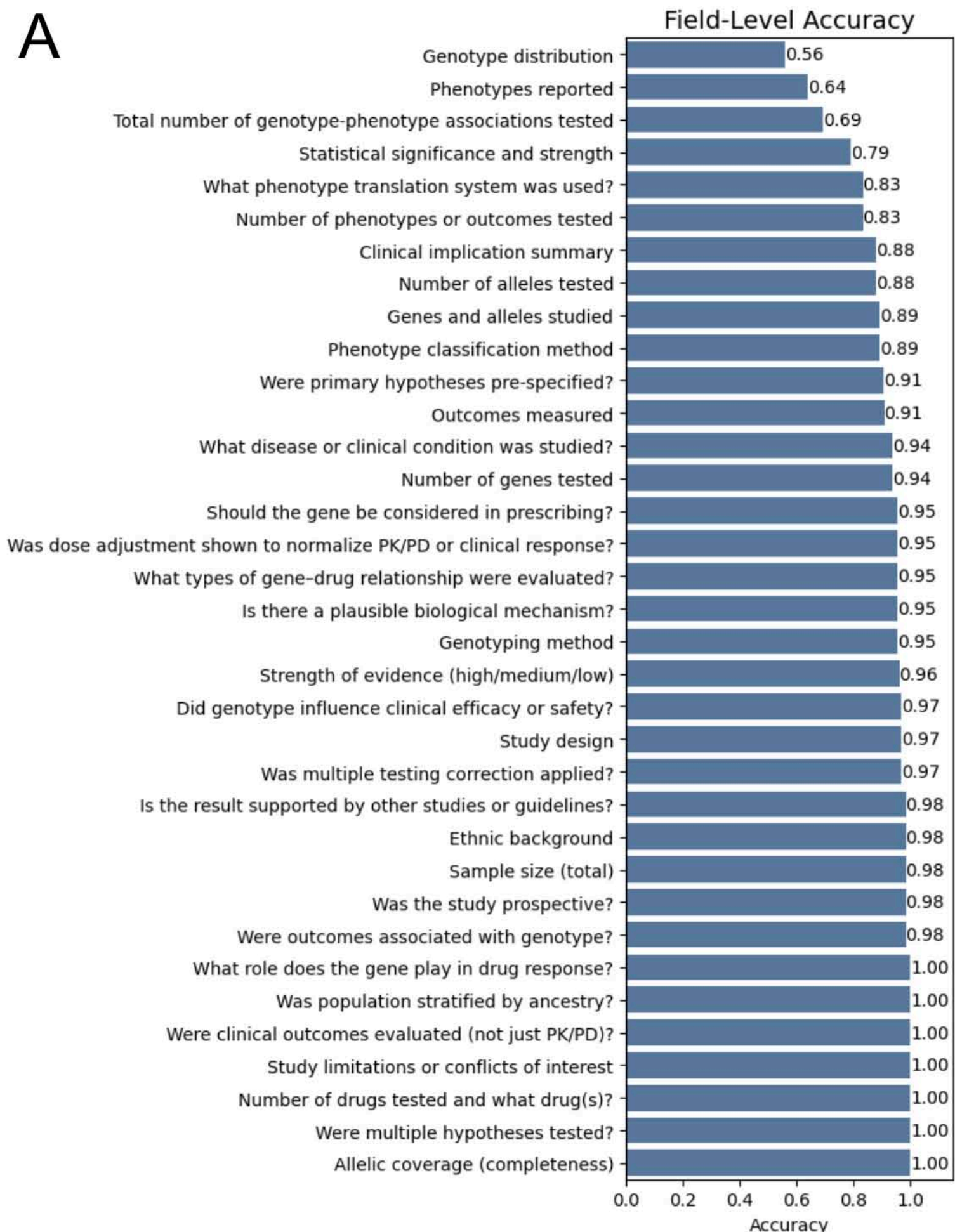
2. Evidence Extraction

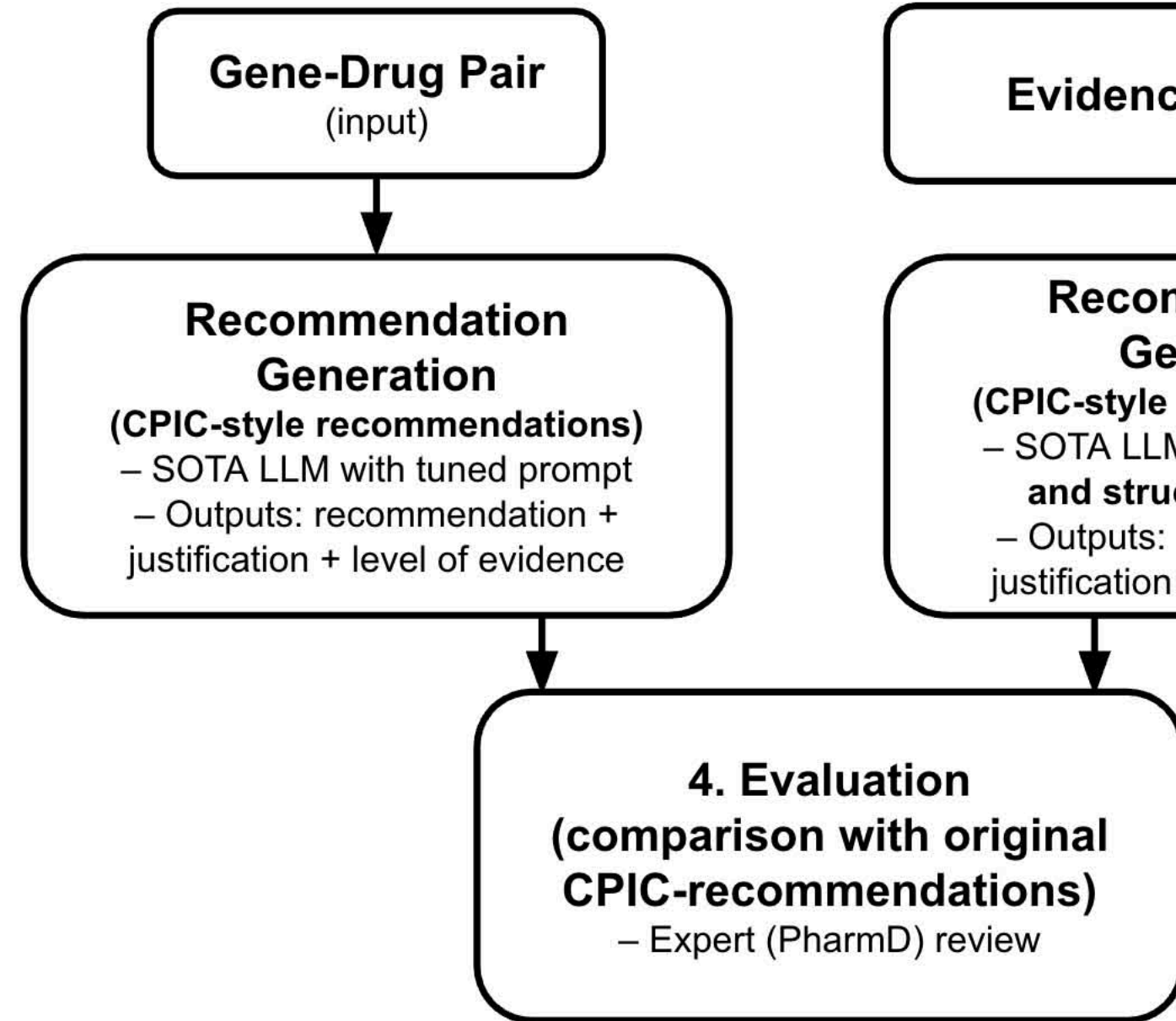
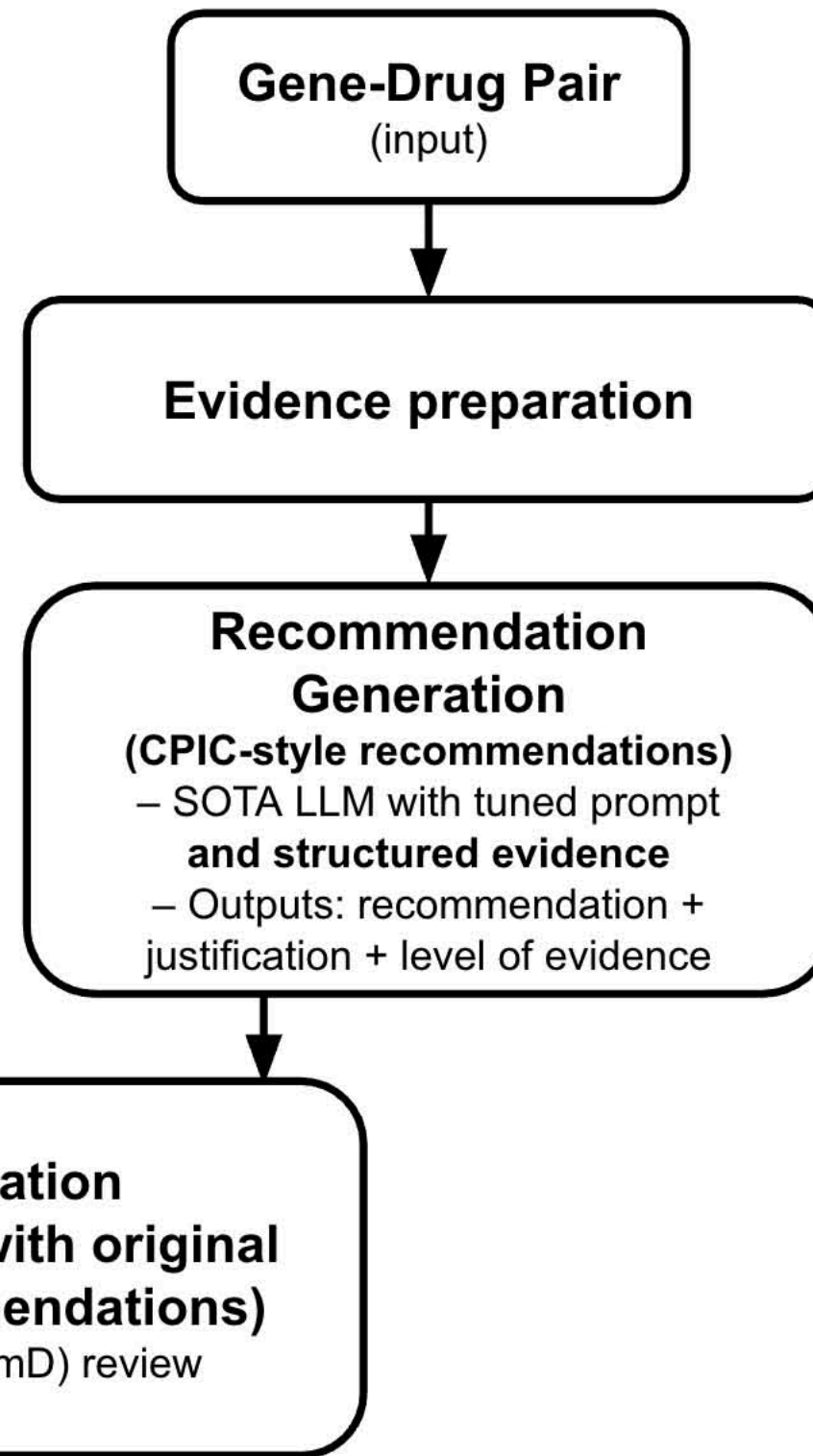
- LLM-based tagging and Q&A
- Structured JSON output

3. Recommendation Generation

(CPIC-style recommendations)

- SOTA LLM with tuned prompt and structured evidence
- Outputs: recommendation + justification + level of evidence



A**Baseline SOTA model****PGxAI-Recommenderr****B****Expert Ratings of Pharmacogenomic Recommendations**